

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - II

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2. Linked List Manipulation:

Write a program to:

a. Create a singly linked list.

Code:

```
class Node:
```

```
    def __init__(self, data):
```

```
        self.data = data
```

```
        self.next = None
```

```
node1 = Node(11)
```

```
node2 = Node(15)
```

```
node3 = Node(7)
```

```
node4 = Node(10)
```

```
node5 = Node(9)
```

```
node1.next = node2
```

```
node2.next = node3
```

```
node3.next = node4
```

```

node4.next = node5
def traver(head):
    curr = head
    while curr:
        print(curr.data, end="=>")
        curr = curr.next
    print("Null")
traver(node1)
print("Himanshu S037")

```

Output:

```

11=>15=>7=>10=>9=>Null
Himanshu S037

```

b. Insert a node at the beginning, end, and at a given position in a linked list.

Code:

```

class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(15)
node3 = Node(7)
node4 = Node(10)
node5 = Node(9)

node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5
global head
head = node1

def traver(head):
    curr = head
    while curr:

```

```

        print(curr.data, end="=>")
        curr = curr.next
    print("Null")

def insert_beg(data):
    global head
    new = Node(data)
    new.next = head
    head = new

def insert_end(data):
    global head
    new = Node(data)

    if head is None:
        head = new
    else:
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new

def ins_pos(pos, data):
    global head
    new = Node(data)

    temp = head
    for i in range(pos-2):
        temp = temp.next
    new.next = temp.next
    temp.next = new

insert_beg(10)
insert_end(25)
ins_pos(2, 17)
traver(head)
print("Himanshu S037")

```

Output:

10=>17=>11=>15=>7=>10=>9=>25=>Null
Himanshu S037

c. Delete a node from a given position in a linked list.

Code:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(15)
node3 = Node(7)
node4 = Node(10)
node5 = Node(9)

node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5

global head
head = node1

def traver(head):
    curr = head
    while curr:
        print(curr.data, end="=>")
        curr = curr.next
    print("Null")
```

```
def insert_beg(data):  
    global head  
    new = Node(data)  
    new.next = node1  
    head = new  
  
def insert_end(data):  
    global head  
    new = Node(data)  
  
    if head is None:  
        head = new  
    else:  
        temp = head  
        while temp.next:  
            temp = temp.next  
        temp.next = new  
  
def ins_pos(pos, data):  
    global head  
    new = Node(data)  
  
    temp = head  
    for i in range(pos-2):  
        temp = temp.next  
    new.next = temp.next  
    temp.next = new  
  
def dele_pos(pos):
```

```

global head

temp = head

for i in range(pos-2):

    temp = temp.next

temp.next = temp.next.next

insert_beg(10)

insert_end(25)

ins_pos(2, 17)

dele_pos(4)

traver(head)

print("Himanshu S037")

```

Output:

```

10=>17=>11=>7=>10=>9=>25=>Null
Himanshu S037

```

Curiosity Questions:

1. Why is inserting a node at the beginning of a singly linked list faster than inserting at the end?
2. What will happen if we don't update the head pointer when inserting a node at the beginning?
3. Why do we need to use a loop when inserting a node at a specific position in the linked list?
4. What would happen if you forgot to update the next pointer during deletion at a given position?
5. What should you check before deleting a node from a position in the linked list? How would you ensure that your program does not crash with invalid positions or empty list operations?